Samarjeet Mohite

Software Engineer – Backend, Platform & Infrastructure
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Summary

Full-stack software engineer with **2+ years** of experience, strongest on the **backend**: scalable APIs, distributed systems and **cloud infrastructure**, plus conversational AI agents and enterprise workflows. Also worked as a **forward deployed engineer**, shipping solutions directly with clients. I like owning things end to end, from design to production.

Education

Indian Institute of Information Technology and Management Gwalior — **Jul 2020 – May 2025**
Integrated B.Tech and M.Tech in Information Technology — *CGPA: 7.9*

Experience

Meraki Labs – Venture Studio — **Mar 2025 – Present**
Software Engineer – shipping AI-native products across ventures — *Bangalore, India*

Fermi AI *– AI Tutoring | Python, FastAPI, PostgreSQL, Celery, Redis, AWS, Terraform, Grafana*

- Ran **AWS ECS** services via **Terraform**: ECR, IAM, secrets, **OIDC deploy roles**, Terraform apply in deploys.
- Stood up **Grafana** on ECS (own IAM, security group, RDS state) with Fleet, Postgres and Redis dashboards.
- Automated incidents: **CloudWatch alarms** to Slack incidents, 5xx/slow-route pulse, **daily SLO report**.
- Cut Postgres connection setups **92%** (64 to 5) when possible connections (**1,050**) exceeded the RDS limit (**450**).
- Shipped **usage metering & billing** on three surfaces, charged per active minute, **no double-charges**.
- Built a **targeting & A/B platform**: FastAPI + Redis API, Celery workers, Python SDK, **MCP** and **OAuth 2.1**.
- Launched **Scan & Ask**, a photo-to-AI-tutor pipeline live on India's **National Digital Library**, **98%** parse accuracy.

Nurix AI *– Voice AI Agents | Python, FastAPI, TypeScript, MySQL, Redis, Lua, SQS, LiteLLM*

- Designed a **Campaign Manager** for outbound calling: **5 lakh+ calls/day**, **500 concurrent**, ∼40 clients.
- Shipped a **telephony re-architecture** on **LiveKit**: SIP trunks, Redis number-to-agent routing, **MCube** provider.
- **Forward-deployed engineer** on **conversational agents** for US/India clients (First Mid, edtech), **$200K+ ARR**.
- Cut **database load 70%** by making blocking calls asynchronous; added **Weighted Fair Scheduling**.
- Engineered a queue orchestrator with per-agent **rate limiting** to trigger outbound calls reliably.
- Integrated a **unified LLM interface** orchestrating model and tool calls across multiple providers.

Projects

GroupMailBox – Lead-capture & Email SaaS | *TypeScript, Chrome MV3, Express, BullMQ*

- Published a **Chrome extension** (v1.0.3) that captures, dedupes and syncs Facebook Group leads to **Google Sheets**.
- Created a **Gmail campaign engine**: **100 emails/min**, 3 retries, **450/day** user cap, signed one-click unsubscribe.
- Added bulk **member-request review** for group admins, verifying each approval and capped at **50 per session**.

Relay – Durable Workflow Engine | *Python, FastAPI, Postgres, OpenTelemetry*

- Built a **Postgres-backed durable execution engine** with journal and replay; **1M** in-flight runs survive any crash.
- Chaos-tested **effectively-once side effects**: killed workers mid-run across **1,000** workflows, exactly **1,000** charges.
- Designed for **1M workflows/day** on a single Postgres (**5k txn/s** at peak) using **SKIP LOCKED** task queues.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Lua
Backend: FastAPI, Node.js, Express.js, Celery, BullMQ, REST APIs, WebSockets, Microservices, Async Processing
Systems: Distributed Systems, System Design, Message Queues, Rate Limiting, Caching, A/B Testing, Observability
AI / LLM: LLM Integration, LiteLLM, MCP, LangGraph, AI Agents, Prompt Evaluation
Databases: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, ClickHouse
Cloud and DevOps: AWS (ECS, ECR, RDS, SQS, S3, IAM), Docker, Kubernetes, Helm, Terraform, CI/CD
Observability: Grafana, CloudWatch, Sentry, OpenTelemetry, Prometheus, SLOs, Alerting
Frontend: React, Next.js, HTML5, CSS

Achievements

- **Codeforces** Specialist (**1562**), **LeetCode** Knight (**1860**), **CodeChef** 4⋆ (**1884**); Codeforces Round 760 rank **38**.
- **Top 1600** of **400k+** teams in **Flipkart Grid 5.0**; **Top 20** of **2000+** students on **GeeksforGeeks** in college.
- General rank **9085** in **IIT JEE Advanced**.